

The Magnitude Mirage: Rethinking Confidence for Reasoning-Intensive Retrieval

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Abstract

Reliable Retrieval-Augmented Generation (RAG) requires detecting when retrieval has failed. Many production RAG systems implement simple similarity thresholding, treating raw retrieval scores as calibrated confidence signals. We expose the *Magnitude Mirage*: as queries transition from simple semantic matching to complex reasoning, neural retrievers often assign high similarity scores to semantically related but constraint-violating documents, causing magnitude thresholding to degrade to near-random abstention performance. To address this without relying on computationally expensive LLM-based evaluators, we conduct a large-scale empirical study evaluating six zero-cost Query Performance Prediction (QPP) metrics across 11 retrieval architectures and 28 datasets. Benchmarking across three cognitive tiers – semantic matching, logical reasoning, and temporal reasoning – we show that score-distribution signals consistently outperform raw magnitude thresholding. In particular, Score Gap ($s_1 - s_k$) and Linearized Score Magnitude and Variance (LSMV) improve abstention AUROC by up to 0.16 across models and reasoning tiers. Because these signals operate solely on the score distribution already produced during retrieval, they do not require additional model inference, retraining, or latency, providing a practical zero-cost replacement for magnitude thresholding in deployed RAG systems.

1 Introduction

Retrieval-Augmented Generation (RAG) systems rely on external retrieval components to provide factual context to large language models (Lewis et al., 2020; Guu et al., 2020; Izacard et al., 2023). The reliability of such systems depends critically on their ability to detect retrieval failures. When the retrieved context does not contain relevant information, downstream generation models may hallucinate or produce incorrect answers (Asai et al.,

2023; Lee et al., 2025; Sun et al., 2025). Consequently, practical RAG pipelines require mechanisms that can predict retrieval success and abstain when confidence is low (Thakur et al., 2023; Asai et al., 2023).

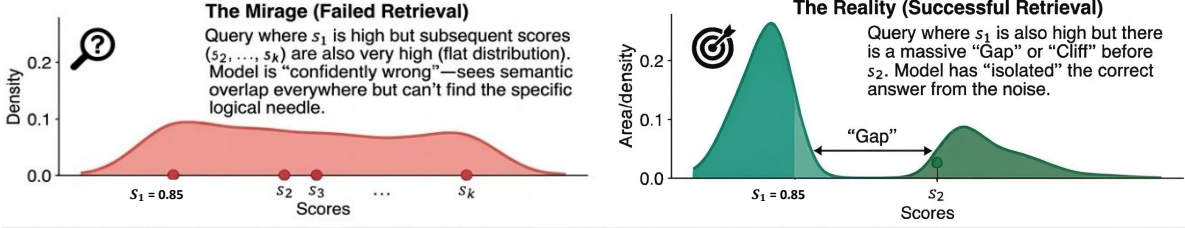
In many industrial RAG frameworks, this decision is often implemented using simple thresholding of raw retrieval scores, a design reflected in widely used toolkits such as LangChain and LlamaIndex (Chase, 2022; Liu, 2022). While computationally inexpensive, this approach implicitly assumes that absolute score magnitudes are comparable across queries and can reliably distinguish successful from unsuccessful retrieval.

However, this assumption breaks as retrieval tasks move beyond simple semantic matching toward queries requiring logical, mathematical, or temporal reasoning. Modern dense retrievers are typically trained to optimize semantic similarity between query and document embeddings (Karpukhin et al., 2020; Thakur et al., 2021). Consequently, a retriever may assign a high similarity score to a document that matches the general semantics of a query while still violating key constraints necessary for correctness. In such cases, the raw retrieval score may appear highly confident even when the retrieved document is irrelevant to the true task requirement. As illustrated in Figure 1, two queries with identical maximum scores can exhibit fundamentally different score distributions. We refer to this phenomenon as the *Magnitude Mirage*: the misleading confidence conveyed by high absolute retrieval scores.

While recent work has explored LLM-based approaches for evaluating retrieved context, such methods introduce substantial latency and computational cost. This motivates revisiting classical Query Performance Prediction (QPP) techniques, which estimate retrieval success directly from the distribution of retrieval scores.

QPP has been extensively studied in classical in-

Section A: Conceptual Score Distributions (The “Why”)



Section B: Threshold Calibration (The “How”)

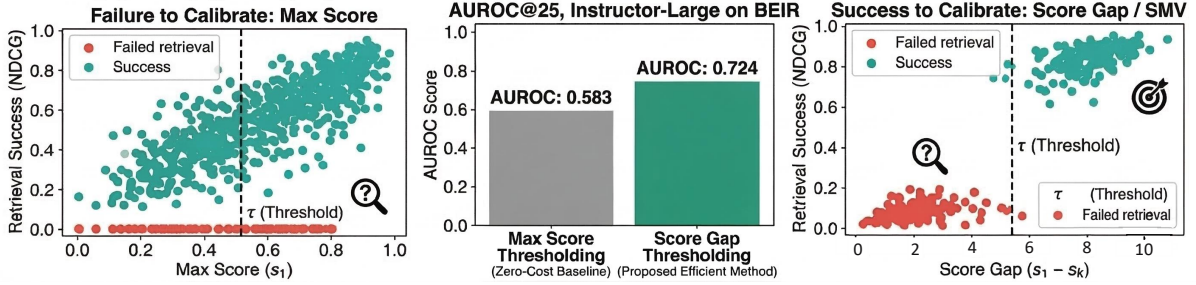


Figure 1: The Magnitude Mirage vs. Reality. **Section A:** Two queries sharing identical top scores ($s_1 = 0.85$) exhibit fundamentally different score distributions. Failed retrieval produces a flat distribution; successful retrieval isolates the correct document via a large Score Gap. **Section B:** MaxScore thresholding cannot reliably distinguish successful from failed retrievals (AUROC: 0.583), while Score Gap thresholding achieves strong calibration (AUROC: 0.724) — AUROC@25 averaged across the evaluated BEIR datasets for Instructor-Large.

formation retrieval, where several zero-cost statistical signals — particularly variance-based measures of score distributions — were proposed to predict query difficulty (Tao and Wu, 2014; He and Ounis, 2004). However, these techniques were originally developed for sparse lexical retrieval models operating on relatively simple semantic retrieval tasks (He and Ounis, 2004; Carmel and Yom-Tov, 2010). Modern RAG pipelines instead rely on neural dense retrievers and are increasingly applied to reasoning-intensive queries involving structured constraints. It therefore remains unclear whether classical QPP signals remain informative in contemporary neural retrieval settings. As a result, most practical RAG systems have largely defaulted to magnitude-based thresholding without systematic evaluation of alternative zero-cost signals.

In this work, we conduct a large-scale empirical study of zero-cost QPP methods in modern neural retrieval pipelines. To assess robustness across task complexity, we evaluate six QPP metrics across eleven retrieval architectures and twenty-eight datasets covering three levels of difficulty: standard semantic retrieval (BEIR subset) (Thakur et al., 2021), complex logical reasoning (BRIGHT) (Su et al., 2025), and temporal reasoning (TEMPO) (Abdallah et al., 2026). We measure predictive performance using AUROC for binary query suc-

cess ($NDCG@k > 0$), which directly corresponds to the abstention decision in RAG systems. In open-domain QA and RAG pipelines, successful retrieval often requires only a single relevant passage within the retrieved context window to enable correct downstream generation (Lewis et al., 2020; Karpukhin et al., 2020). In addition, we report Pearson and Spearman correlations with $NDCG@k$ to evaluate continuous prediction quality.

Our results reveal significant limitations of magnitude-based thresholding. As illustrated in Figure 1, absolute score thresholds exhibit unstable behavior even on standard semantic retrieval tasks and degrade substantially on reasoning-intensive benchmarks. In contrast, variance-based QPP signals — particularly Score Gap ($s_1 - s_k$) and Linearized Score Magnitude and Variance (LSMV) — provide consistently stronger indicators of retrieval success across models and datasets.

Our contributions are as follows:

- We identify and systematically characterize the *Magnitude Mirage*, showing that raw retrieval score thresholds are unreliable indicators of retrieval success, particularly for reasoning-intensive queries.
- We present a large-scale evaluation of six zero-

139	cost QPP metrics across eleven retrieval archi-	such as Normalized Query Commitment (NQC)	188
140	tectures and twenty-eight datasets spanning	and Score Magnitude and Variance (SMV), along	189
141	semantic, logical, and temporal reasoning re-	with variants such as our Linearized SMV (LSMV),	190
142	trieval tasks.	quantify the dispersion of top-ranked retrieval	191
143		scores, based on the intuition that confident re-	192
144	• We demonstrate that variance-based QPP met-	trieval results produce strong separation between	193
145	rics provide a simple and computationally free	relevant documents and the corpus background dis-	194
146	alternative to magnitude thresholding, offer-	tribution. These techniques correlate strongly with	195
147	ing more reliable abstention signals for mod-	retrieval effectiveness in classical information re-	196
148	ern RAG pipelines.	trieval systems, particularly sparse lexical models	197
149		such as BM25 (He and Ounis, 2004; Tao and Wu,	198
150	2 Related Work and Background	2014).	199
151	2.1 Retrieval Reliability and RAG Abstention		200
152	Retrieval-Augmented Generation (RAG) systems	Recent work has revisited classical QPP methods	201
153	rely on external retrievers to provide factual context	in the context of neural dense retrieval. Faggioli	202
154	to language models. The presence of irrelevant or	et al. (2023) and others have investigated whether	203
155	misleading documents in the retrieved context can	score-distribution statistics remain predictive for	204
156	substantially degrade downstream generation, even	modern dual-encoder and dense retrievers. How-	205
157	when only a single distracting passage is included	ever, prior evaluations largely focus on standard se-	206
158	(Cuconasu et al., 2024).	matic retrieval benchmarks. The behavior of QPP	207
159	A growing body of work leverages generative	signals under reasoning-intensive retrieval tasks —	208
160	models themselves to evaluate retrieved documents,	where documents may be semantically similar yet	209
161	commonly referred to as <i>LLM-as-a-judge</i> pipelines	logically incorrect — remains largely unexplored.	
162	(Gu et al., 2024; Liu et al., 2025). While effective,		
163	these approaches incur substantial computational	2.3 Reasoning-Intensive Retrieval	210
164	cost and latency, limiting their applicability in high-	Recent retrieval benchmarks such as BRIGHT and	211
165	throughput or real-time systems. Other methods	TEMPO introduce tasks that require logical or tem-	212
166	attempt to improve retrieval reliability through re-	poral reasoning beyond semantic similarity. In	213
167	triever alignment or reinforcement learning (Zhou	these settings, semantically similar documents may	214
168	and Chen, 2026), but these require retraining and	violate critical constraints necessary for correctness.	215
169	do not provide lightweight, inference-time abstention	This raises new challenges for retrieval confidence	216
170	signals.	estimation, as high similarity scores do not neces-	217
171	In contrast, our work investigates lightweight	sarily correspond to correct retrieval.	218
172	confidence estimation methods that operate directly		219
173	on the retriever’s score distribution at inference	Despite the growing importance of reasoning-	220
174	time. Such zero-cost signals do not require ad-	intensive retrieval, relatively little work has exam-	221
175	ditional computational overhead such as model	ined how retrieval confidence signals behave in	222
176	calls or retraining, making them suitable for high-	such settings. Our work addresses this gap by	223
177	throughput RAG pipelines.	systematically evaluating classical QPP metrics	224
178		across both traditional semantic retrieval datasets	225
179	2.2 Query Performance Prediction	and modern reasoning-intensive retrieval datasets.	226
180	Query Performance Prediction (QPP) aims to es-		227
181	timate retrieval success for a given query without	3 Methodology	228
182	relying on human relevance judgments. Tradition-		229
183	ally, QPP methods are divided into <i>pre-retrieval</i>	3.1 Problem Formulation	230
184	and <i>post-retrieval</i> approaches. Pre-retrieval meth-	We formalize the standard Retrieval-Augmented	231
185	ods estimate query difficulty using query statis-	Generation (RAG) first-stage retrieval process.	232
186	tics such as term specificity or corpus distributions,	Given a user query q and a corpus of documents	233
187	whereas post-retrieval methods analyze properties	$C = \{d_1, d_2, \dots, d_N\}$, a retrieval model R maps	234
	of the retrieved document list itself.	the query and corpus to a ranked list of the top- k	
	Among post-retrieval methods, variance-based	documents	
	metrics have proven particularly effective. Methods		
		$D_k = [d_{(1)}, d_{(2)}, \dots, d_{(k)}],$	

where each retrieved document $d_{(i)}$ is associated with a retrieval score $s_i \in \mathbb{R}$, such that

$$s_1 \geq s_2 \geq \dots \geq s_k.$$

Let $S_k = \{s_1, s_2, \dots, s_k\}$ denote the set of top- k retrieval scores.

The goal of Query Performance Prediction (QPP) is to estimate the retrieval effectiveness for a given query q without access to relevance judgments. Among the various classes of QPP methods, we focus specifically on *zero-cost* QPP predictors that operate solely on the score distribution S_k . Formally, a QPP function

$$f(S_k) \rightarrow c_q$$

maps the score distribution to a scalar confidence estimate c_q that approximates the true retrieval effectiveness of the query.

This confidence estimate can then be used in two ways. First, it can serve as a binary decision signal for retrieval success (e.g., abstention in RAG pipelines). Second, it can act as a continuous predictor of retrieval quality, enabling correlation analysis between predicted confidence and observed effectiveness metrics such as $NDCG@k$.

3.2 The Magnitude Mirage

An increasingly common heuristic for zero-cost RAG abstention relies on absolute magnitude thresholding. In practice, RAG frameworks such as LangChain and LlamaIndex apply fixed similarity thresholds to filter retrieved documents, passing only those where $s_i > \tau$ to the generator. This design implicitly assumes that raw retrieval scores are calibrated confidence signals and that a high score reliably indicates a relevant document regardless of query type. Under this assumption, the maximum score s_1 serves as a natural proxy for overall retrieval confidence: if the top-ranked document scores highly, the retrieval is considered successful.

As retrieval tasks shift from semantic matching to reasoning-oriented queries, the calibration of absolute retrieval scores degrades. As shown in Figure 1, even when two queries share an identical top score, MaxScore thresholding cannot reliably distinguish successful from failed retrieval — achieving AUROC of only 0.583 compared to 0.724 for Score Gap on the same queries. This failure reflects a fundamental mismatch between what magnitude measures (semantic proximity) and what abstention requires (constraint satisfaction).

3.3 Variance-Based QPP Metrics

To address the potential miscalibration of raw score magnitudes, we evaluate a suite of distribution-based Query Performance Prediction (QPP) metrics that analyze the structure of retrieval score distributions. The underlying intuition is that successful retrieval often produces a more structured score distribution in which relevant documents—potentially multiple—receive consistently higher scores than the broader corpus background. In contrast, unsuccessful retrieval may produce flatter score distributions where many documents receive similar scores. Variance-based QPP metrics attempt to capture these differences in score dispersion. We formalize the metrics utilized in our evaluation as follows:

Absolute Magnitude (Baseline): The maximum retrieval score:

$$MaxScore = s_1,$$

Score Gap (Top-to-Tail Dropoff): The difference between the top-ranked score and the k -th score:

$$Gap@k = s_1 - s_k$$

where s_k is the lowest score within the top- k retrieved documents.

Top- k Standard Deviation: The standard deviation of the top- k documents, where $\mu = \frac{1}{k} \sum_{i=1}^k s_i$ is the mean of S_k :

$$\sigma@k = \sqrt{\frac{1}{k} \sum_{i=1}^k (s_i - \mu)^2}$$

Normalized Query Commitment (NQC): Adapting the original metric proposed by Shtok et al. (2012) for dense retrievers, we measure the standard deviation normalized by the absolute mean of the top- k scores:

$$NQC@k = \frac{\sigma(S_k)}{\frac{1}{k} \sum_{i=1}^k |s_i|}$$

Maximum Iterative Standard Deviation (σ_{max}): Proposed by Pérez-Iglesias and Araujo (2010), this metric computes the maximum standard deviation across iteratively expanding top- i subsets:

$$\sigma_{max}@k = \max_{i \in [2, k]} \sigma(S_i)$$

Linearized Score Magnitude and Variance (LSMV): The original SMV metric (Tao and Wu, 2014) relied on logarithmic scaling to tame the long-tail distributions of sparse lexical models like BM25. We measure a linearized variant adapted for neural score distributions:

$$LSMV@k = s_1 \times \sigma(S_k)$$

4 Experimental Setup

4.1 Datasets and Cognitive Tiers

To evaluate the robustness of QPP signals across varying retrieval difficulty, we benchmark metrics across 28 datasets grouped into three cognitive tiers capturing different forms of reasoning constraints:

- **Semantic Matching (BEIR):** We utilize four representative datasets from the BEIR benchmark: SciFact, FiQA, NFCorpus, SciDocs. These datasets evaluate standard semantic retrieval tasks.
- **Logical Reasoning (BRIGHT):** We evaluate on the BRIGHT benchmark, which introduces queries requiring logical constraints, multi-hop reasoning, and structured reasoning conditions across twelve datasets. In these tasks, simple semantic similarity is often insufficient to identify correct documents.
- **Temporal Reasoning (TEMPO):** We include the TEMPO benchmark consisting of twelve datasets to evaluate retrieval under explicit and implicit temporal reasoning constraints. In this setting, documents may share high semantic similarity with a query while still violating required time conditions.

4.2 Retrieval Models

To ensure our findings generalize across retrieval paradigms, we evaluate QPP metrics across a diverse set of retrieval architectures.

- **Sparse Lexical Models:** We include BM25 (Robertson and Zaragoza, 2009) as a classical lexical retrieval baseline.
- **Dense Bi-Encoders:** We evaluate several widely used dense retrieval models, including BGE (Xiao et al., 2024), E5 (Wang et al., 2022), Contriever (Izacard et al., 2021), SBERT (Reimers and Gurevych, 2019), SFR (Meng et al., 2024), Qwen (Chu et al., 2024), and Instructor-Large (Su et al., 2023).

- **Reasoning-Augmented Encoders:** We additionally include models explicitly trained or fine-tuned to handle structured or reasoning-intensive retrieval tasks, including Diver (Long et al., 2025), RaDeR (Das et al., 2025), and ReasonIR (Shao et al., 2025).

This diverse model suite enables evaluation across both classical and modern neural retrieval architectures.

4.3 Evaluation Protocol

For each query, we extract the raw retrieval score distribution S_k from each model over the top- k retrieved documents.

We evaluate QPP metrics across multiple context window sizes commonly used in RAG pipelines:

$$k \in \{5, 10, 25, 50\}.$$

Binary Retrieval Success We define retrieval success as the presence of at least one relevant document within the retrieved set:

$$NDCG@k > 0.$$

This definition reflects the practical requirement that at least one relevant document be present in the retrieval context to enable correct downstream generation (Lewis et al., 2020; Karpukhin et al., 2020).

Evaluation Metrics We evaluate the predictive quality of QPP signals using three complementary statistical measures:

- **AUROC (Area Under the Receiver Operating Characteristic Curve).** This serves as our primary evaluation metric and simulates the RAG abstention decision. AUROC measures the ability of a QPP signal to distinguish between successful and failed retrieval instances across all possible decision thresholds, rather than relying on a fixed confidence threshold.
- **Pearson Correlation Coefficient (r).** Measures the linear correlation between the predicted QPP confidence value and the observed retrieval effectiveness ($NDCG@k$).
- **Spearman Rank Correlation (ρ).** Measures the monotonic relationship between predicted confidence and retrieval effectiveness, providing robustness to non-linear scaling effects.

All metrics are computed per dataset and reported as macro-averages across datasets within each benchmark.

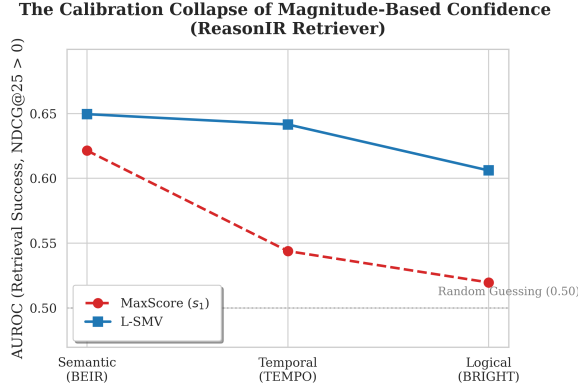


Figure 2: LSMV vs. MaxScore AUROC@25 illustrating systematic degradation of magnitude-based confidence as reasoning difficulty increases.

4.4 Implementation Details

Experiments were conducted on a compute node equipped with 4× NVIDIA H100-80GB GPUs, 512GB system memory, and AMD EPYC processors using PyTorch and HuggingFace Transformers. All dense models use FP16 inference with official Huggingface checkpoints.

5 Results

5.1 The Magnitude Mirage Across Cognitive Tiers

Table 1 presents AUROC@25 scores for binary retrieval success across all three cognitive tiers. The results reveal a consistent pattern across all models: raw score magnitude (MaxScore) becomes an unreliable abstention signal for reasoning-intensive retrieval tasks, and its failure is systematic rather than incidental. Figure 2 illustrates this collapse for ReasonIR: MaxScore AUROC degrades steadily across cognitive tiers, approaching random performance on logical retrieval, while LSMV maintains stable discrimination.

On standard semantic retrieval (BEIR), MaxScore performs adequately for some models — BGE (.749) and SFR (.754) achieve reasonable discrimination — but already show instability for reasoning-augmented encoders such as ReasonIR (.621) and RaDeR (.666). The degradation becomes dramatic on reasoning-intensive tasks. On TEMPO, MaxScore collapses to near-random performance for Qwen (.536) and ReasonIR (.544). On BRIGHT, the failure is universal: *every* model tested produces MaxScore AUROC values in the range .520–.611, providing essentially no actionable abstention signal. This confirms the core claim of the Magni-

tude Mirage: neural retrievers confidently assign high scores to logically or temporally incorrect documents, and the raw magnitude of those scores carries no reliable information about retrieval success.

5.2 Variance-Based Metrics Restore Calibration

In contrast, variance-based QPP signals — particularly Score Gap and LSMV — consistently recover meaningful abstention performance across all three tiers. On BEIR, Score Gap and LSMV improve over MaxScore for nearly every model, with gains of up to .158 AUROC (Instructor-Large: .583 → .741). On TEMPO, where MaxScore degrades most severely, Score Gap achieves gains of up to .125 AUROC (Qwen: .536 → .661). On BRIGHT, Score Gap and LSMV improve over MaxScore by margins ranging from .060 to .091 across dense models.

Critically, these gains are consistent across both dense bi-encoders and reasoning-augmented encoders, suggesting that variance-based metrics capture a structural property of confident retrieval that persists across architectures: when a retriever correctly identifies a relevant document under complex constraints, that document statistically separates from the background distribution, producing a measurable score gap.

5.3 Comparison Across All Six QPP Metrics

Full results for all six QPP metrics are provided in the Appendix (Table 2). Here we summarize the key observations.

First, all variance-based alternatives uniformly and substantially outperform MaxScore across all settings. Across dense models on BEIR, the improvement from switching to any variance-based metric ranges from .048 (SFR) to .158 (Instructor-Large), while the spread *among* variance-based metrics for the same models is at most .017. On BRIGHT, where MaxScore AUROC is compressed into the narrow band .525–.571 across dense models, variance-based metrics expand this to .590–.653 — a recovery of meaningful discrimination from near-random baselines. The pattern is consistent: the gain from abandoning magnitude exceeds the differences among distributional alternatives by a factor of roughly 5–10×, confirming that the core contribution is the shift from magnitude to distributional estimation, rather than the specific choice of distributional statistic.

Model	BEIR (Semantic)			TEMPO (Temporal)			BRIGHT (Logical)		
	Max	Gap	LSMV	Max	Gap	LSMV	Max	Gap	LSMV
<i>Sparse Models</i>									
BM25	.701	.695	.700	.472	.526	.510	.611	.622	.624
<i>Dense Models</i>									
BGE	.749	.805	.810	.626	.701	.705	.538	.608	.608
E5	.736	.777	.787	.648	.709	.718	.552	.628	.633
Contriever	.683	.727	.733	.591	.705	.707	.560	.624	.626
SBERT	.718	.766	.772	.638	.691	.720	.525	.605	.590
SFR	.754	.790	.802	.649	.671	.690	.553	.623	.622
Qwen	.698	.787	.788	.536	.661	.654	.571	.653	.650
Instructor-Large	.583	.724	.741	.563	.664	.669	.534	.619	.608
<i>Reasoning Encoders</i>									
Diver-Retriever	.665	.788	.789	.582	.666	.684	.543	.605	.606
RaDeR	.666	.740	.752	.570	.646	.665	.560	.645	.640
ReasonIR	.621	.626	.650	.544	.638	.642	.520	.611	.606

Table 1: Comparison of zero-cost QPP metrics across three cognitive tiers. We report AUROC@25 scores for binary retrieval success ($NDCG@25 > 0$). Max = s_1 , Gap = $s_1 - s_{25}$, and LSMV represents the Linearized Score Magnitude and Variance metric. The baseline magnitude column is shaded gray; best result per row–tier is **bolded**.

Second, Score Gap and LSMV emerge as the most consistent pair across tiers, though no single metric dominates universally. LSMV achieves the best or joint-best AUROC across the majority of BEIR configurations, while Score Gap provides stronger results on several BRIGHT and TEMPO settings. NQC offers complementary strength for sparse retrieval: BM25 on TEMPO shows NQC (.602) substantially outperforming Score Gap (.526), reflecting its stability under the heavier-tailed score distributions of term-frequency models. The key practical implication is that any variance-based signal provides reliable abstention where magnitude fails — the specific choice is secondary.

5.4 Effect of Retrieval Window Size

The comprehensive per-benchmark results (Tables 4, 5, 6 in the Appendix) show that the advantages of variance-based metrics are stable across $k \in \{5, 10, 25, 50\}$. AUROC improvements over MaxScore are observed at every cutoff tested, indicating that the Magnitude Mirage is not an artifact of any particular context window size. There is a modest tendency for Score Gap and LSMV to widen their advantage as k increases, consistent with the intuition that larger retrieval windows provide richer score distributions from which variance signals can be estimated.

5.5 Correlation with Continuous Retrieval Effectiveness

Pearson and Spearman correlations with $NDCG@k$ (reported in the Appendix tables) confirm the same ordering. MaxScore shows weak or near-zero linear correlation with retrieval effectiveness on reasoning-intensive tasks (e.g., Pearson $r < .03$ for Qwen and BM25 on TEMPO at $k = 25$), while Score Gap and LSMV achieve correlations up to .40 on the same benchmarks. Spearman correlations exhibit the same pattern, confirming the relationship is monotonic rather than an artifact of linear scaling assumptions.

For example, Score Gap achieves Pearson $r = .467$ vs. MaxScore $r = .328$ for Qwen on BEIR, $r = .401$ vs. $r = .193$ on BRIGHT and $r = .303$ vs. $r = .028$ on TEMPO. Full correlations across all models and cutoffs are reported in Appendix Table 3.

6 Analysis and Discussion

6.1 Why Magnitude Fails in Reasoning Retrieval

Modern dense retrievers are trained with contrastive objectives that reward semantic proximity, without explicitly encoding logical, mathematical, or temporal constraints. Consequently, the top-ranked score s_1 reflects only semantic compatibility — not constraint satisfaction. Multiple documents may achieve similarly high scores while violating critical task conditions, causing magnitude-based thresholding to fail to distinguish correct

from incorrect retrieval. This explains the systematic MaxScore collapse observed on BRIGHT and TEMPO.

6.2 Why Variance Signals Capture Retrieval Confidence

Variance-based metrics operate on score distribution *structure* rather than absolute magnitude. When a retriever correctly identifies a relevant document under complex constraints, that document often separates from the corpus background, producing a steeply declining score distribution with measurable gaps. As shown in Figure 3, on BRIGHT’s TheoremQA subset a vertical MaxScore threshold produces near-random class separation, while a horizontal Score Gap threshold partitions successful from failed retrievals.

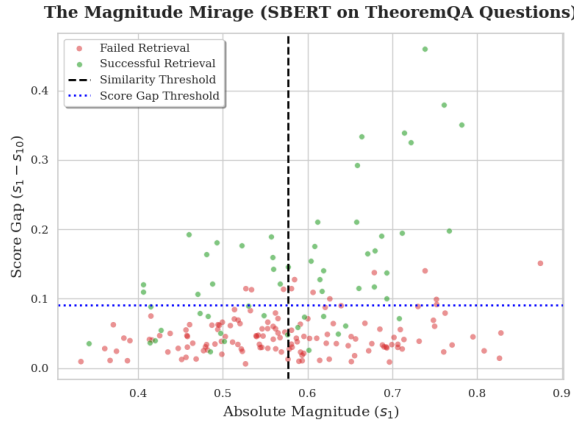


Figure 3: SBERT on BRIGHT TheoremQA. Red = failed retrieval, Green = successful retrieval. A vertical MaxScore threshold provides near-random separation; a horizontal Score Gap threshold partitions the two classes.

This separation is not an artifact of corpus sparsity: reasoning-intensive benchmarks contain comparable or higher relevant document density than semantic retrieval tasks (BRIGHT averages 4–5 relevant documents per query, TEMPO averages 2–4, vs. 1–5 for BEIR), confirming that the observed score separation reflects retrieval confidence rather than a statistical property of the relevance annotations. Failed retrieval produces flatter distributions where many semantically similar documents receive comparable scores. This structural separation is architecture-agnostic: it emerges consistently across dense bi-encoders and reasoning-augmented encoders alike, explaining the uniform gains of Score Gap and LSMV.

6.3 Implications for RAG System Design

Our results suggest a simple revision to standard RAG abstention: replace absolute magnitude thresholding with a variance-based confidence signal. Across dense retrieval architectures and all three cognitive tiers, variance-based signals consistently outperform MaxScore, with improvements larger than the differences among the variance metrics themselves. This indicates that the primary benefit comes from abandoning magnitude-based confidence estimation rather than from the specific choice of variance metric.

In practice, Score Gap and LSMV provide strong default choices given their consistent performance. Because these metrics operate solely on the top- k score distribution already produced during retrieval, they can be integrated into existing RAG pipelines at *zero cost*, without additional model calls, retraining, or latency.

7 Conclusion

We identify the *Magnitude Mirage* — the systematic failure of absolute retrieval score thresholds as confidence signals under logical and temporal reasoning constraints — and show it consistently affects neural retrievers across architectures. The fix is simple: classical variance-based QPP metrics, specifically Score Gap and LSMV, restore reliable abstention performance at zero additional cost by measuring score distribution structure rather than raw magnitude.

Our broader finding is that the shift from magnitude to distributional estimation matters more than which distributional statistic is chosen — all six variance-based metrics substantially outperform MaxScore. This suggests that the RAG community’s reliance on magnitude thresholding is not a calibration problem to be tuned, but a structural mismatch between what retrievers optimize and what abstention requires. As retrieval tasks continue to grow in reasoning complexity, this mismatch is likely to widen, making distribution-aware confidence estimation an increasingly critical component of reliable RAG system design.

Limitations

First, our evaluation measures retrieval confidence at the retriever stage rather than the full end-to-end RAG pipeline. While at least one relevant document in the retrieved context is a necessary condition for correct generation, it is not sufficient

— downstream models may still fail even when retrieval succeeds. Validating whether variance-based abstention translates to measurable reductions in hallucination remains an important direction for future work.

Second, our experiments are limited to publicly available benchmarks. Real-world retrieval systems may exhibit different score calibration properties due to domain-specific corpora, proprietary query distributions, or custom retriever training procedures not represented here.

Third, we define binary retrieval success as $NDCG@k > 0$, reflecting the presence of at least one relevant document. Stricter success criteria — requiring higher relevance grades or multiple relevant documents — may produce different calibration behaviors and warrant separate investigation.

Finally, while we establish that variance-based signals substantially outperform magnitude thresholding, we do not characterize the absolute performance ceiling for zero-cost abstention. Whether the remaining gap relative to computationally expensive approaches such as cross-encoder reranking or LLM-based verification can be closed without additional inference remains an open question.

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Model	BEIR (Semantic)						TEMPO (Temporal)						BRIGHT (Logical)					
	Max	Gap	Std	NQC	MIS	LSMV	Max	Gap	Std	NQC	MIS	LSMV	Max	Gap	Std	NQC	MIS	LSMV
<i>Sparse</i>																		
BM25	.701	.695	.694	.663	.688	.700	.472	.526	.529	.602	.530	.510	.611	.622	.626	.611	.623	.624
<i>Dense</i>																		
BGE	.749	.805	.805	.795	.805	.810	.626	.701	.698	.696	.703	.705	.538	.608	.611	.615	.608	.608
E5	.736	.777	.780	.770	.778	.787	.648	.709	.719	.719	.708	.718	.552	.628	.636	.638	.630	.633
Contriever	.683	.727	.728	.701	.730	.733	.591	.705	.715	.708	.701	.707	.560	.624	.625	.618	.619	.626
SBERT	.718	.766	.770	.713	.769	.772	.638	.691	.716	.696	.694	.720	.525	.605	.600	.608	.597	.590
SFR	.754	.790	.795	.782	.791	.802	.649	.671	.688	.684	.675	.690	.553	.623	.625	.626	.624	.622
Qwen	.698	.787	.785	.771	.783	.788	.536	.661	.661	.660	.656	.654	.571	.653	.652	.650	.646	.650
Instructor-L	.583	.724	.740	.738	.718	.741	.563	.664	.671	.670	.660	.669	.534	.619	.609	.610	.611	.608
<i>Reasoning</i>																		
Diver-Retriever	.665	.788	.790	.789	.786	.789	.582	.666	.681	.681	.663	.684	.543	.605	.606	.605	.599	.606
RaDeR	.666	.740	.748	.743	.744	.752	.570	.646	.668	.670	.658	.665	.560	.645	.640	.640	.644	.640
ReasonIR	.621	.626	.624	.583	.627	.650	.544	.638	.652	.653	.641	.642	.520	.611	.616	.626	.616	.606

Table 2: AUROC@25 for all six zero-cost QPP metrics across three cognitive tiers. **Max** = MaxScore (s_1), **Gap** = Score Gap ($s_1 - s_{25}$), **Std** = Top- k Standard Deviation, **NQC** = Normalized Query Commitment, **MIS** = Maximum Iterative Standard Deviation, **LSMV** = Linearized Score Magnitude and Variance. Baseline magnitude column is shaded gray; best result per row–tier is **bolded**.

Model	k	BEIR		BRIGHT		TEMPO	
		Max	Gap	Max	Gap	Max	Gap
BM25	5	.330	.307	.149	.177	.029	.095
	10	.344	.334	.175	.195	.019	.106
	25	.347	.353	.191	.208	-.000	.070
	50	.360	.366	.206	.224	.017	.095
BGE	5	.391	.460	.131	.256	.198	.210
	10	.391	.505	.131	.270	.198	.272
	25	.391	.538	.131	.274	.198	.302
	50	.391	.541	.131	.256	.198	.314
E5	5	.350	.443	.142	.233	.192	.227
	10	.357	.470	.159	.284	.201	.270
	25	.363	.495	.157	.302	.211	.311
	50	.364	.495	.153	.285	.218	.323
Contriever	5	.286	.416	.119	.275	.101	.240
	10	.296	.430	.127	.298	.105	.281
	25	.297	.442	.129	.308	.129	.312
	50	.301	.439	.136	.297	.147	.328
SBERT	5	.356	.448	.102	.244	.193	.225
	10	.367	.483	.121	.266	.185	.276
	25	.366	.495	.114	.273	.216	.326
	50	.365	.493	.119	.281	.241	.366
SFR	5	.374	.440	.171	.266	.162	.200
	10	.375	.467	.169	.300	.181	.255
	25	.377	.492	.173	.312	.194	.300
	50	.376	.493	.172	.298	.195	.315
Qwen	5	.329	.410	.190	.325	.013	.217
	10	.335	.451	.188	.366	.018	.257
	25	.328	.467	.193	.401	.028	.303
	50	.327	.470	.197	.404	.023	.328
Instructor-Large	5	.203	.346	.144	.274	.046	.181
	10	.201	.399	.139	.298	.069	.219
	25	.193	.437	.139	.314	.086	.261
	50	.191	.453	.138	.292	.090	.260
Diver-Retriever	5	.261	.445	.139	.295	.098	.209
	10	.262	.468	.150	.342	.084	.239
	25	.257	.482	.159	.347	.106	.274
	50	.255	.476	.164	.339	.118	.285
RaDeR	5	.227	.402	.130	.312	.052	.136
	10	.242	.446	.131	.355	.075	.223
	25	.247	.462	.136	.358	.087	.261
	50	.246	.456	.139	.353	.093	.280
ReasonIR	5	.170	.287	.117	.268	.003	.142
	10	.177	.294	.126	.298	.001	.174
	25	.186	.297	.120	.309	.019	.198
	50	.187	.283	.123	.303	.025	.211

Table 3: Pearson correlation between QPP signals and retrieval effectiveness across benchmarks and cutoffs. Max denotes MaxScore (s_1) and Gap denotes Score Gap ($s_1 - s_k$).

Model	k	MaxScore			ScoreGap			Top- k Std			NQC			MaxIterStd			LSMV		
		AUC	r	ρ	AUC	r	ρ	AUC	r	ρ	AUC	r	ρ	AUC	r	ρ	AUC	r	ρ
Sparse Models																			
BM25	5	.696	.330	.340	.694	.307	.341	.694	.305	.341	.671	.268	.307	.694	.189	.343	.698	.240	.345
	10	.712	.344	.351	.709	.334	.361	.705	.323	.350	.676	.282	.322	.703	.194	.355	.711	.245	.354
	25	.701	.347	.348	.695	.353	.357	.694	.329	.338	.663	.284	.307	.688	.209	.349	.700	.252	.349
	50	.700	.360	.355	.691	.366	.358	.687	.327	.331	.646	.272	.287	.680	.220	.346	.696	.263	.349
Dense Models																			
BGE	5	.749	.391	.397	.755	.460	.487	.754	.456	.487	.742	.444	.470	.753	.349	.487	.763	.455	.498
	10	.749	.391	.397	.786	.505	.534	.787	.500	.526	.776	.488	.514	.783	.388	.529	.793	.497	.533
	25	.749	.391	.397	.805	.538	.558	.805	.514	.530	.795	.510	.524	.805	.423	.557	.810	.510	.534
	50	.749	.391	.397	.808	.541	.561	.793	.482	.500	.783	.483	.496	.805	.425	.556	.800	.483	.507
E5	5	.701	.350	.354	.754	.443	.472	.755	.440	.471	.748	.429	.461	.753	.355	.470	.759	.440	.479
	10	.722	.357	.361	.759	.470	.487	.766	.465	.481	.758	.455	.472	.763	.386	.488	.770	.466	.487
	25	.736	.363	.367	.777	.495	.501	.780	.456	.459	.770	.452	.456	.778	.412	.492	.787	.459	.466
	50	.739	.364	.367	.787	.495	.498	.776	.408	.417	.766	.410	.418	.786	.413	.482	.784	.416	.427
Contriever	5	.663	.286	.279	.705	.416	.414	.704	.414	.413	.686	.394	.389	.703	.353	.413	.713	.412	.422
	10	.672	.296	.293	.714	.430	.435	.717	.421	.431	.693	.398	.404	.716	.365	.434	.724	.418	.435
	25	.683	.297	.303	.727	.442	.453	.728	.408	.431	.701	.392	.409	.730	.376	.453	.733	.402	.429
	50	.694	.301	.308	.728	.439	.452	.717	.365	.399	.688	.360	.383	.726	.375	.447	.726	.364	.401
SBERT	5	.701	.356	.355	.724	.448	.456	.727	.447	.458	.686	.379	.394	.725	.374	.457	.744	.449	.481
	10	.713	.367	.366	.741	.483	.496	.740	.474	.488	.690	.397	.423	.738	.404	.492	.753	.469	.497
	25	.718	.366	.371	.766	.495	.513	.770	.454	.478	.713	.399	.421	.769	.418	.506	.772	.443	.477
	50	.726	.365	.373	.782	.493	.513	.779	.402	.441	.699	.363	.378	.789	.406	.503	.781	.400	.447
SFR	5	.722	.374	.375	.761	.440	.478	.761	.432	.477	.751	.418	.462	.759	.329	.476	.767	.432	.487
	10	.738	.375	.375	.773	.467	.493	.775	.458	.479	.763	.444	.467	.771	.363	.486	.782	.458	.489
	25	.754	.377	.379	.790	.492	.506	.795	.451	.458	.782	.445	.455	.791	.392	.494	.802	.453	.467
	50	.744	.376	.378	.796	.493	.503	.786	.409	.416	.774	.411	.418	.796	.393	.487	.793	.417	.428
Qwen	5	.691	.329	.312	.730	.410	.423	.727	.403	.420	.712	.379	.399	.723	.327	.416	.738	.408	.434
	10	.711	.335	.314	.764	.451	.462	.764	.444	.453	.747	.421	.433	.759	.371	.456	.771	.447	.462
	25	.698	.328	.316	.787	.467	.472	.785	.434	.435	.771	.420	.422	.783	.392	.468	.788	.435	.442
	50	.716	.327	.315	.802	.470	.472	.787	.397	.400	.769	.392	.389	.803	.395	.466	.794	.403	.412
Instructor-Large	5	.593	.203	.191	.669	.346	.309	.668	.343	.306	.667	.340	.303	.662	.298	.297	.669	.345	.308
	10	.592	.201	.189	.695	.399	.364	.704	.413	.378	.703	.411	.376	.690	.356	.358	.705	.416	.380
	25	.583	.193	.187	.724	.437	.418	.740	.453	.435	.738	.450	.431	.718	.399	.410	.741	.455	.438
	50	.582	.191	.186	.724	.453	.435	.737	.451	.438	.734	.448	.432	.723	.415	.433	.739	.454	.444
Reasoning Encoders																			
Diver-Retriever	5	.667	.261	.267	.756	.445	.470	.755	.439	.464	.752	.434	.463	.753	.341	.464	.756	.435	.465
	10	.666	.262	.265	.775	.468	.489	.772	.455	.474	.771	.453	.475	.769	.371	.479	.773	.450	.472
	25	.665	.257	.267	.788	.482	.502	.790	.442	.458	.789	.445	.462	.786	.397	.490	.789	.437	.458
	50	.675	.255	.265	.805	.476	.494	.810	.396	.420	.808	.404	.426	.815	.395	.486	.810	.393	.422
RaDeR	5	.634	.227	.227	.707	.402	.405	.707	.399	.404	.704	.396	.401	.705	.319	.402	.708	.396	.405
	10	.641	.242	.241	.724	.446	.444	.732	.450	.448	.729	.449	.446	.727	.372	.448	.733	.444	.447
	25	.666	.247	.252	.740	.462	.461	.748	.437	.439	.743	.442	.441	.744	.390	.462	.752	.432	.438
	50	.683	.246	.253	.758	.456	.461	.761	.387	.403	.753	.397	.409	.760	.382	.458	.765	.384	.405
ReasonIR	5	.601	.170	.177	.628	.287	.248	.625	.280	.241	.607	.244	.212	.625	.258	.244	.634	.288	.255
	10	.605	.177	.193	.619	.294	.253	.615	.284	.240	.583	.232	.190	.614	.263	.245	.630	.295	.261
	25	.621	.186	.207	.626	.297	.262	.624	.268	.239	.583	.202	.184	.627	.260	.258	.650	.288	.271
	50	.619	.187	.205	.621	.283	.245	.614	.230	.202	.569	.161	.143	.621	.247	.239	.641	.260	.246

Table 4: Comprehensive zero-cost QPP evaluation across all target cutoffs ($k \in \{5, 10, 25, 50\}$) for the **BEIR** benchmark. Predictive power is evaluated using AUROC (AUC), Pearson (r), and Spearman (ρ).

Model	k	MaxScore			ScoreGap			Top-k Std			NQC			MaxIterStd			LSMV		
		AUC	r	ρ	AUC	r	ρ	AUC	r	ρ	AUC	r	ρ	AUC	r	ρ	AUC	r	ρ
Sparse Models																			
BM25	5	.597	.149	.153	.613	.177	.180	.615	.178	.182	.601	.177	.163	.612	.132	.178	.617	.153	.184
	10	.602	.175	.177	.612	.195	.209	.618	.199	.217	.598	.206	.191	.612	.138	.207	.615	.164	.209
	25	.611	.191	.206	.622	.208	.238	.626	.197	.244	.611	.214	.229	.623	.127	.241	.624	.161	.237
	50	.604	.206	.220	.615	.224	.250	.612	.206	.247	.600	.218	.235	.611	.136	.255	.611	.177	.241
Dense Models																			
BGE	5	.538	.131	.088	.583	.256	.167	.584	.260	.167	.584	.261	.168	.586	.240	.170	.583	.258	.166
	10	.538	.131	.088	.602	.270	.195	.603	.266	.196	.606	.270	.201	.603	.248	.197	.600	.261	.192
	25	.538	.131	.088	.608	.274	.202	.611	.247	.201	.615	.256	.208	.608	.244	.199	.608	.243	.197
	50	.538	.131	.088	.604	.256	.193	.596	.204	.173	.601	.216	.182	.602	.227	.187	.593	.203	.170
E5	5	.537	.142	.090	.608	.233	.202	.606	.237	.200	.607	.234	.200	.604	.195	.196	.605	.236	.199
	10	.546	.159	.110	.621	.284	.240	.617	.291	.235	.617	.290	.236	.613	.243	.227	.617	.288	.234
	25	.552	.157	.122	.628	.302	.259	.636	.305	.275	.638	.310	.279	.630	.260	.263	.633	.301	.269
	50	.544	.153	.115	.625	.285	.253	.616	.256	.245	.621	.263	.253	.621	.237	.251	.614	.254	.242
Contriever	5	.565	.119	.101	.644	.275	.204	.643	.269	.201	.641	.259	.199	.643	.236	.202	.643	.271	.202
	10	.575	.127	.119	.654	.298	.246	.652	.298	.243	.648	.295	.239	.651	.255	.242	.654	.294	.244
	25	.560	.129	.120	.624	.308	.251	.625	.287	.257	.618	.295	.252	.619	.262	.248	.626	.279	.254
	50	.526	.136	.118	.629	.297	.241	.634	.240	.240	.641	.252	.245	.629	.243	.240	.624	.239	.236
SBERT	5	.536	.102	.075	.617	.244	.189	.616	.236	.188	.617	.231	.187	.615	.215	.186	.612	.233	.184
	10	.544	.121	.095	.619	.266	.218	.615	.255	.213	.617	.250	.213	.615	.228	.213	.610	.249	.206
	25	.525	.114	.079	.605	.273	.228	.600	.256	.224	.608	.258	.234	.597	.230	.221	.590	.245	.208
	50	.519	.119	.087	.601	.281	.242	.596	.243	.233	.605	.253	.245	.591	.231	.233	.584	.231	.216
SFR	5	.568	.171	.130	.615	.266	.218	.616	.268	.217	.615	.262	.216	.612	.227	.212	.614	.270	.214
	10	.550	.169	.119	.623	.300	.248	.623	.307	.249	.626	.305	.252	.619	.260	.242	.623	.304	.249
	25	.553	.173	.136	.623	.312	.272	.625	.310	.275	.626	.315	.279	.624	.263	.272	.622	.305	.269
	50	.551	.172	.135	.628	.298	.269	.614	.263	.249	.618	.272	.256	.622	.240	.257	.611	.260	.244
Qwen	5	.598	.190	.180	.663	.325	.305	.659	.321	.298	.654	.310	.290	.658	.277	.296	.662	.324	.304
	10	.594	.188	.183	.667	.366	.341	.665	.375	.340	.661	.363	.333	.663	.329	.335	.667	.378	.342
	25	.571	.193	.180	.653	.401	.360	.652	.407	.356	.650	.396	.353	.646	.363	.353	.650	.407	.353
	50	.583	.197	.190	.637	.404	.362	.630	.380	.345	.626	.375	.344	.628	.362	.352	.632	.382	.343
Instructor-Large	5	.581	.144	.121	.649	.274	.235	.651	.273	.238	.651	.270	.237	.650	.241	.236	.652	.274	.238
	10	.545	.139	.095	.639	.298	.247	.638	.288	.241	.638	.286	.241	.635	.258	.240	.637	.289	.240
	25	.534	.139	.093	.619	.314	.253	.609	.296	.231	.610	.297	.233	.611	.266	.237	.608	.296	.231
	50	.525	.138	.090	.594	.292	.233	.584	.248	.207	.586	.251	.210	.588	.240	.222	.584	.249	.207
Reasoning Encoders																			
Diver-Retriever	5	.567	.139	.135	.625	.295	.271	.622	.288	.265	.619	.288	.260	.622	.234	.264	.623	.285	.267
	10	.572	.150	.154	.633	.342	.316	.629	.337	.313	.628	.338	.312	.627	.283	.307	.631	.331	.313
	25	.543	.159	.159	.605	.347	.316	.606	.336	.308	.605	.341	.309	.599	.298	.303	.606	.329	.308
	50	.554	.164	.163	.628	.339	.313	.624	.294	.278	.624	.301	.281	.631	.285	.298	.622	.290	.277
RaDeR	5	.566	.130	.125	.655	.312	.295	.654	.309	.293	.652	.306	.292	.651	.253	.290	.653	.307	.294
	10	.550	.131	.117	.653	.355	.329	.651	.357	.328	.651	.359	.327	.648	.294	.322	.650	.352	.326
	25	.560	.136	.133	.645	.358	.334	.640	.349	.324	.640	.355	.327	.644	.294	.331	.640	.342	.323
	50	.548	.139	.142	.660	.353	.356	.650	.309	.322	.653	.318	.327	.650	.266	.335	.647	.303	.319
ReasonIR	5	.564	.117	.117	.627	.268	.237	.628	.268	.240	.627	.269	.239	.626	.227	.237	.627	.260	.237
	10	.559	.126	.124	.624	.298	.265	.625	.303	.269	.623	.304	.267	.623	.254	.262	.622	.290	.261
	25	.520	.120	.101	.611	.309	.273	.616	.308	.281	.626	.321	.293	.616	.255	.281	.606	.288	.266
	50	.534	.123	.118	.616	.303	.291	.620	.272	.271	.623	.295	.285	.618	.243	.280	.614	.255	.260

Table 5: Comprehensive zero-cost QPP evaluation across all target cutoffs ($k \in \{5, 10, 25, 50\}$) for the **BRIGHT** benchmark. Predictive power is evaluated using AUROC (AUC), Pearson (r), and Spearman (ρ).

Model	k	MaxScore			ScoreGap			Top- k Std			NQC			MaxIterStd			LSMV		
		AUC	r	ρ	AUC	r	ρ	AUC	r	ρ	AUC	r	ρ	AUC	r	ρ	AUC	r	ρ
Sparse Models																			
BM25	5	.508	.029	.040	.563	.095	.115	.561	.086	.112	.573	.119	.126	.559	.039	.110	.555	.045	.103
	10	.504	.019	.030	.581	.106	.148	.571	.105	.130	.608	.175	.190	.563	.052	.124	.548	.055	.097
	25	.472	-.000	-.015	.526	.070	.091	.529	.062	.090	.602	.174	.209	.530	.026	.096	.510	.014	.059
	50	.505	.017	.024	.559	.095	.126	.560	.071	.118	.642	.209	.258	.560	.048	.128	.526	.028	.061
Dense Models																			
BGE	5	.626	.198	.226	.629	.210	.260	.633	.204	.260	.628	.206	.252	.632	.116	.251	.638	.195	.269
	10	.626	.198	.226	.673	.272	.326	.670	.266	.320	.664	.266	.310	.670	.156	.320	.672	.253	.323
	25	.626	.198	.226	.701	.302	.364	.698	.317	.349	.696	.320	.347	.703	.187	.357	.705	.303	.361
	50	.626	.198	.226	.708	.314	.371	.704	.311	.361	.704	.316	.360	.714	.190	.366	.705	.299	.368
E5	5	.619	.192	.217	.668	.227	.306	.663	.217	.301	.660	.218	.296	.662	.109	.300	.667	.206	.306
	10	.645	.201	.237	.708	.270	.362	.709	.272	.356	.706	.273	.353	.702	.129	.341	.710	.257	.354
	25	.648	.211	.242	.709	.311	.382	.719	.344	.394	.719	.348	.399	.708	.145	.377	.718	.327	.380
	50	.625	.218	.251	.670	.323	.384	.681	.358	.387	.677	.363	.388	.661	.147	.373	.686	.342	.386
Contriever	5	.585	.101	.121	.649	.240	.264	.638	.234	.250	.633	.243	.244	.636	.148	.248	.639	.213	.251
	10	.588	.105	.127	.692	.281	.322	.685	.279	.306	.671	.285	.296	.683	.171	.306	.688	.256	.310
	25	.591	.129	.159	.705	.312	.377	.715	.317	.368	.708	.323	.359	.701	.191	.358	.707	.295	.358
	50	.596	.147	.176	.694	.328	.381	.698	.327	.357	.689	.333	.357	.690	.198	.350	.694	.306	.354
SBERT	5	.636	.193	.209	.657	.225	.270	.651	.223	.260	.641	.209	.250	.649	.130	.256	.657	.212	.269
	10	.624	.185	.199	.673	.276	.316	.673	.279	.313	.659	.263	.299	.665	.148	.302	.677	.259	.326
	25	.638	.216	.236	.691	.326	.371	.716	.370	.408	.696	.343	.369	.694	.202	.367	.720	.344	.408
	50	.654	.241	.261	.708	.366	.412	.743	.403	.437	.717	.380	.405	.705	.232	.400	.747	.381	.428
SFR	5	.625	.162	.215	.627	.200	.264	.616	.184	.244	.612	.186	.238	.615	.090	.242	.623	.172	.257
	10	.647	.181	.239	.675	.255	.338	.675	.256	.340	.667	.257	.332	.664	.111	.311	.680	.239	.344
	25	.649	.194	.236	.671	.300	.362	.688	.331	.376	.684	.333	.372	.675	.127	.344	.690	.311	.368
	50	.622	.195	.237	.660	.315	.369	.698	.359	.385	.692	.365	.390	.679	.137	.363	.694	.340	.373
Qwen	5	.483	.013	-.005	.643	.217	.268	.643	.216	.270	.640	.235	.260	.641	.127	.265	.640	.186	.267
	10	.503	.018	.023	.648	.257	.303	.646	.266	.288	.650	.286	.288	.645	.157	.285	.641	.231	.281
	25	.536	.028	.022	.661	.303	.302	.662	.304	.289	.660	.318	.297	.656	.182	.277	.654	.270	.273
	50	.510	.023	.002	.648	.328	.320	.654	.331	.299	.659	.348	.328	.643	.215	.282	.635	.293	.259
Instructor-Large	5	.556	.046	.097	.624	.181	.253	.619	.177	.251	.623	.180	.252	.623	.092	.258	.620	.170	.251
	10	.553	.069	.127	.617	.219	.282	.611	.224	.269	.615	.227	.276	.611	.116	.273	.611	.216	.267
	25	.563	.086	.125	.664	.261	.332	.671	.280	.324	.670	.284	.320	.660	.129	.319	.669	.271	.320
	50	.557	.090	.140	.664	.260	.335	.705	.294	.358	.708	.298	.361	.664	.117	.343	.698	.285	.349
Reasoning Encoders																			
Diver-Retriever	5	.563	.098	.124	.619	.209	.282	.619	.210	.279	.617	.213	.270	.618	.110	.276	.618	.199	.277
	10	.558	.084	.104	.650	.239	.325	.657	.257	.323	.656	.262	.318	.648	.122	.319	.656	.242	.323
	25	.582	.106	.125	.666	.274	.351	.681	.309	.349	.681	.314	.353	.663	.143	.350	.684	.293	.357
	50	.557	.118	.135	.626	.285	.342	.643	.304	.323	.640	.309	.327	.625	.147	.330	.643	.291	.330
RaDeR	5	.553	.052	.078	.601	.136	.197	.600	.139	.193	.597	.142	.190	.600	.068	.195	.602	.133	.195
	10	.543	.075	.083	.630	.223	.267	.631	.234	.256	.632	.237	.257	.625	.139	.248	.631	.225	.258
	25	.570	.087	.100	.646	.261	.307	.668	.299	.311	.670	.304	.315	.658	.175	.299	.665	.287	.309
	50	.619	.093	.102	.650	.280	.345	.653	.315	.334	.635	.322	.328	.648	.180	.315	.659	.303	.320
ReasonIR	5	.531	.003	.041	.630	.142	.260	.629	.142	.255	.632	.161	.258	.630	.040	.255	.627	.103	.249
	10	.547	.001	.052	.638	.174	.281	.633	.190	.275	.641	.211	.295	.632	.042	.262	.635	.138	.263
	25	.544	.019	.064	.638	.198	.311	.652	.246	.322	.653	.268	.338	.641	.048	.301	.642	.183	.298
	50	.581	.025	.072	.685	.211	.338	.698	.270	.350	.685	.291	.361	.675	.048	.328	.698	.206	.304

Table 6: Comprehensive zero-cost QPP evaluation across all target cutoffs ($k \in \{5, 10, 25, 50\}$) for the **TEMPO** benchmark. Predictive power is evaluated using AUROC (AUC), Pearson (r), and Spearman (ρ).